Pricing a Chokepoint:
A Synthetic-Control Estimate of the Brent Crude Premium from the 2026 Strait of Hormuz Disruption

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Abstract

We estimate the causal effect of the 28 February 2026 Strait of Hormuz disruption on the price of Brent crude. Because Brent is a single global benchmark with no natural control unit, the central problem is the construction of a credible counterfactual price path. We build that counterfactual with the *synthetic control method* (SCM), using a pool of 19 macro-financial donor series (metals, agriculturals, equities, foreign exchange, rates, and volatility) selected to share Brent's exposure to common global factors while staying off the oil-supply causal path. To guard against any single modelling choice driving the result, the synthetic is formed as the median of an *ensemble* of five estimators with different inductive biases (convex SCM, augmented SCM, elastic net, gradient-boosted trees, and Bayesian ridge). The design is first validated on a known disruption, the 24 February 2022 Russian invasion of Ukraine, where it recovers a price path consistent with the documented \$95→\$130 move, and is then applied out-of-sample to Hormuz. Over the post-event window to mid-June 2026 we estimate an average Brent premium of roughly 56 % (ensemble median; inter-quartile range 51 % to 57 %). The estimate survives an in-space placebo test (Brent ranks first among all placebo units; permutation p at the attainable floor), a leave-one-donor-out check, a contamination-bias sign analysis, and a comparison against naïve univariate counterfactuals. We frame the estimate as a reduced-form, data-driven complement to the structural scenario estimates produced by energy institutions, and discuss its use as an input to strategic-reserve, monetary, and fiscal decisions.

Contents

1 Introduction

The Strait of Hormuz is the narrowest binding constraint in the global oil system. Roughly one-fifth of world petroleum-liquids consumption transits the lane between Iran and Oman, and no fully redundant bypass exists: the Saudi East–West and UAE Habshan–Fujairah pipelines together carry only a fraction of the seaborne volume, leaving the remainder dependent on the maritime route. A disruption to this chokepoint, whether by blockade, mining, military exchange, or insurance-driven withdrawal of shipping, is the single largest unhedged supply-side risk in the oil market. Periodic near-misses (the 1980s tanker war, the 2019 Abqaiq strike, repeated tensions through the mid-2020s) produced transient risk-premium spikes without an actual closure. The 28 February 2026 event is the realisation of that long-anticipated tail risk, and it raises a sharp empirical question:

> *What is the causal effect of the 2026 Strait of Hormuz disruption on the price of Brent crude over a policy-relevant post-event window?*

Answering this requires the *counterfactual* Brent path, meaning what the price would have done absent the disruption, against which the observed path is compared. The construction of that counterfactual, not the comparison itself, is the methodological core of this thesis.

Why the magnitude matters. A credible price-impact estimate is a direct input to decisions that are routinely taken under acute uncertainty. Emergency strategic-reserve releases (as coordinated by the IEA after the 2022 invasion) are sized against an implied price displacement; the monetary-policy choice to "look through" an oil shock or to tighten depends on its size and expected persistence; fiscal stabilisation in oil-importing economies with fuel subsidies scales with the price move; and sovereign hedging programmes (e.g. Mexico's annual oil hedge) price disruption scenarios against exactly such magnitudes. In each application the binding constraint is the quality of the price-impact estimate.

Why synthetic control. Brent is a single, globally traded benchmark. This rules out the standard observational toolkits that need many treated and control units (difference-in-differences), a running variable around a cutoff (regression discontinuity), or a credible instrument. Three feasible options remain: an event study on abnormal returns, which gives a short-window magnitude but no multi-month counterfactual path; a structural vector autoregression with identified oil-supply shocks (Kilian, 2009; Baumeister and Hamilton, 2019), which requires identification restrictions with no standard precedent for a chokepoint event; and the synthetic control method (Abadie et al., 2010; Abadie, 2021), which builds a weighted basket of donor series whose pre-event co-movement with Brent is high and projects it forward as the counterfactual. We use SCM because it (i) produces a full counterfactual *path* rather than a point estimate, the object the policy levers above act on; (ii) makes its identifying assumptions auditable through the donor pool and a per-event treatment catalogue; and (iii) needs no structural stance on whether the move is supply, risk premium, or expectations: the estimand is a reduced-form total effect, which is what most applications need. We target the average treatment effect on the treated (ATT), the mean gap between observed and counterfactual Brent over the post-event window, because it integrates over the whole path (including any reversion) rather than over a single peak day.

Why two events. Applying SCM to Hormuz alone has a hard limitation: a reviewer presented with one number cannot tell a sound estimate from an artefact of donor pool, pre-window, model class, and regularisation interacting. We therefore calibrate the design on the **2022 Russian**

invasion of Ukraine, a large, well-documented Brent disruption ($95→$130 within two weeks), and verify that the same machinery recovers that magnitude before applying it out-of-sample to Hormuz. The two events also share a 19-donor pool, which enables the strongest single robustness test available with two events, a *cross-event weight transfer* in which Russia-fitted weights are applied to the Hormuz window to check that the factor structure SCM relies on is regime-stable.

Contributions and use of MSc tools. Substantively, we produce a data-driven Brent-price ATT for the Hormuz disruption that complements the structural scenario estimates of energy institutions; the two answer different questions (price implied by an assumed barrel shortfall vs. price gap implied by observed cross-asset co-movement), so their agreement is informative. Methodologically, we extend the SCM-on-price design of Becker et al. (2021) from a regulated retail-fuel market with cross-country donors to a globally traded crude benchmark with *cross-asset* donors, under an exogenous geopolitical treatment, and we treat the validity of that extension as something to be demonstrated rather than assumed. Throughout we lean on tools from the MSc: the causal-inference framework of potential outcomes and SCM; the machine-learning regularisation and tree-ensemble methods that make up the estimator ensemble; time-series cross-validation and structural-break testing; and permutation/randomisation inference. Each tool is explained where it is used.

The remainder proceeds as follows. Section 2 reviews the literature by *topic*. Section 3 sets out the conceptual framework: a factor model of Brent, the estimand, the identifying assumptions, and the ensemble. Section 4 describes the data and reports summary statistics and the donor audit. Section 5 states the empirical strategy and the validation battery. Section 6 reports results for both events. Section 7 concludes. Appendices collect the robustness grid, the Bayesian-ridge modelling note, and the mathematics of the structural-break tests.

2 Related Literature

We organise the review around *issues* rather than individual articles, because the contribution of this thesis sits at the intersection of three otherwise separate strands: how causal counterfactuals are built for aggregate units; how the oil-price effect of geopolitical shocks is usually measured; and how inference is conducted when there is a single treated unit and a short post-period.

2.1 Building counterfactuals for a single aggregate unit

The synthetic control method originates with Abadie and Gardeazabal (2003) and Abadie et al. (2010), who construct a treated unit's counterfactual as a convex combination of untreated donors matched on pre-treatment outcomes, and formalise permutation (placebo) inference for the single-treated-unit case. Abadie et al. (2015) extend the approach and its placebo logic to comparative politics, and the survey of Abadie (2021) codifies the practical requirements: a long, shock-free pre-window; donors that are clean of the treatment (the SCM analogue of SUTVA); and a good pre-treatment fit as a precondition for credibility. A methodological literature then relaxes the convex-hull restriction. The *augmented* SCM of Ben-Michael et al. (2021) adds a ridge bias-correction that allows limited extrapolation when the treated unit lies outside the donor hull; Doudchenko and Imbens (2016) embed SCM, difference-in-differences, and regression in a common regularised-regression family that permits negative and unconstrained weights; Athey et al. (2021) recast the problem as matrix completion. A recurring caution is that convex SCM weights are *non-unique* when donors are correlated (Abadie and L'Hour, 2021): many weight vectors yield near-identical pre-period fit, so individual weights should not be over-interpreted. We exploit this entire toolbox by running an ensemble that spans the convex, augmented, and regularised-regression cases, and we use the non-uniqueness result to interpret cross-model weight dispersion correctly.

2.2 What SCM has, and has not, been applied to

Most peer-reviewed SCM applications estimate effects on *non-price* aggregate outcomes such as GDP, trade, and employment, including under discrete geopolitical shocks: Gharehgozli (2017) estimates the GDP cost of sanctions on Iran, and Mancini et al. (2024) the trade effect of the 2022 Russia–Ukraine war. SCM on a *daily market price* is rarer; the closest mechanical precedent is Becker et al. (2021), who apply SCM to Austrian retail fuel prices with other European countries as donors. That design differs from ours in three respects: the treatment is an endogenously chosen regulation rather than an exogenous shock, the outcome is an administered retail price rather than a traded benchmark, and the donors are cross-country rather than cross-asset. Juárez et al. (2024) use SCM with an oil-price shock as treatment but on regional GDP, not on a price. The intersection we occupy, SCM applied to a crude-benchmark *price* for an exogenous *geopolitical* disruption, has limited direct precedent, and that is the methodological gap the thesis addresses.

2.3 Oil prices and geopolitical risk: the incumbent methods

The dominant empirical approach to oil-price effects of macro and geopolitical events is the structural VAR with identified supply, demand, and inventory shocks (Kilian, 2009; Kilian and Murphy, 2014; Baumeister and Hamilton, 2019); Hamilton (2009) provides the narrative anchor for the 2007–08 episode. These decompose price changes into structural components but require identification restrictions, and a chokepoint event has no standard precedent for what those restrictions should be. A parallel strand uses the Geopolitical Risk index of Caldara and Iacoviello (2022) as a regressor in oil-price models; that answers the *average* response of prices to a unit increase in geopolitical risk pooled across many events, which is a different object from the counterfactual effect of one specific disruption. We therefore use the GPR index only as a descriptive overlay, never as a donor, because it spikes endogenously to the very events we study. Short-window event studies on abnormal returns (Brown and Warner, 1985) round out the incumbent set; they give a magnitude but not the multi-month path that policy needs.

2.4 Inference, pretesting, and short post-periods

The single-treated-unit setting makes inference non-standard. Permutation/placebo tests (Abadie et al., 2010, 2015) are the field standard, with formal validity resting on an approximate-symmetry (exchangeability) assumption (Canay et al., 2017; Lehmann and Romano, 2005); Hahn and Shi (2017) show that for convex SCM this effectively requires Gaussian errors, which bears directly on how far the placebo p-values can be pushed for our nonlinear ensemble members. Chen and Yan (2023) formalise the in-time placebo into a "mixed" placebo test with a permutation p-value. On the pre-trend side, Roth (2022) shows that pretests for parallel trends have low power at applied sample sizes and can manufacture false confidence, and Rambachan and Roth (2023) propose bounding rather than testing pre-trend violations; we adopt their report-and-interpret stance toward pre-period diagnostics instead of imposing pass/fail thresholds the SCM literature does not use. For multiple testing across donors we use Benjamini–Hochberg control (Benjamini and Hochberg, 1995). Closely related to our setting, Ham and Miratrix (2024) analyse the bias–variance trade-off of matching on pre-treatment outcomes before a difference-in-differences analysis and show that the bias reduction is proportional to outcome *reliability*, that is, how tightly the outcome is coupled to the latent covariates, while regression-to-the-mean injects an offsetting bias. Although their setting is DiD, the lesson transfers directly to SCM: a good pre-period fit can mask, rather than remove, contamination, and the value of a donor depends on how reliably it loads on the common factor. We use exactly this reliability logic to split donors by pre-window factor R^2 when we sign the contamination bias (Section 6).

2.5 Tools imported from time series and machine learning

The estimator ensemble and its validation draw on standard ML and time-series machinery, each used as a means to a causal end. Out-of-sample fit is assessed by expanding-window (rolling-origin) cross-validation, the canonical scheme for autocorrelated series (Hyndman and Athanasopoulos, 2018; Bergmeir and Benítez, 2012), read through the bias–variance lens of Hastie et al. (2009). The ensemble members are gradient-boosted trees (Friedman, 2001; Chen and Guestrin, 2016), the elastic net (Zou and Hastie, 2005), and a Bayesian ridge regression that stands in for the regression component of the Bayesian structural time-series model of Brodersen et al. (2015). Hyperparameter selection is disciplined against the selection-bias ("winner's curse") of Cawley and Talbot (2010). Structural-break and regime-stability diagnostics use Bai and Perron (1998, 2003), the I(0)/I(1)-robust trend-break test of Harvey et al. (2009), distance correlation (Székely et al., 2007), and the KPSS statistic (Kwiatkowski et al., 1992); the donor-cleanliness battery uses the known-break test of Chow (1960) in a permutation form (Lehmann and Romano, 2005) rather than the unknown-break test of Andrews (1993).

2.6 The focal event and its institutional comparison set

Because the Hormuz disruption is recent, no peer-reviewed journal study of it can yet exist; early policy-relevant analysis lives in institutional work: the Dallas Fed study of the 2026 Iran war's inflation impact (Federal Reserve Bank of Dallas, 2026), the Kiel Policy Brief on the cost of closing the Strait (Kiel Institute for the World Economy, 2026), an Oxford Institute for Energy Studies modelling note (Oxford Institute for Energy Studies, 2026), the World Bank Commodity Markets Outlook (World Bank, 2026), and the IEA Oil Market Reports (International Energy Agency, 2026). These share a structure: assume a barrels-per-day shortfall, propagate it through a structural model, and output a price path. None applies synthetic control to the observed price series; that is the comparison set against which we position our estimate.

3 Conceptual Framework

A good data-science analysis needs an explicit view of the data-generating process. This section states that view, derives the estimand and the identifying assumptions from it, and explains why an *ensemble* of estimators is the right proxy when the functional form of the counterfactual is unknown.

3.1 A factor model of Brent

Write log Brent as the sum of a part spanned by global macro-financial factors common to many assets and an oil-specific part:

$$p_t^{\text{Brent}} = \underbrace{\lambda' F_t}_{\text{common global factors}} + \underbrace{o_t}_{\text{oil-specific (supply, chokepoint, geopolitics)}} + \varepsilon_t. \tag{1}$$

F_t collects the five drivers that move Brent in normal times (the global industrial cycle, the US dollar, real interest rates, global risk appetite, and China demand), each with non-oil proxies. o_t is the object of interest: the oil-supply and chokepoint component that a Hormuz disruption shifts. Each donor j has its own factor representation $X_{jt} = \gamma_j' F_t + u_{jt}$ with, by construction, *zero loading on* o_t: a clean donor is exposed to the same global factors as Brent but is not physically routed through oil supply. A synthetic control is a weighted donor basket $\hat{p}_t = \sum_j w_j X_{jt}$. If the weights reproduce Brent's factor loadings, $\sum_j w_j \gamma_j \approx \lambda$, then in the pre-window, where o_t is small and stable, the basket tracks Brent. When a chokepoint shock fires, o_t jumps for Brent but not for the basket, the two decouple, and the decoupling *is* the estimated premium. Donor selection is therefore the economic exercise of spanning F_t while excluding o_t, that is, choosing identification over fit. This is why the energy complex (WTI, refined products, petrocurrencies), despite the highest raw correlation with Brent, is excluded by design: it loads on the very o_t we are trying to measure.

3.2 Potential outcomes and the estimand

Let Y_t^N be Brent's potential log-price absent the event and Y_t^I under it, with treatment beginning at T_0. The treated-unit effect is $\tau_t = Y_t^I - Y_t^N$ for $t \geq T_0$. With a single treated unit the synthetic supplies the missing $\hat{Y}_t^N = \sum_j w_j X_{jt}$, and the estimated effect is the gap $\hat{\tau}_t = Y_t^I - \hat{Y}_t^N$. We report it in percent, $\text{gap}_t = 100\,[\exp(Y_t^I - \hat{Y}_t^N) - 1]$, and summarise the post-window by the *average treatment effect on the treated*,

$$\widehat{\text{ATT}} \;=\; \frac{1}{|W|} \sum_{t \in [T_0,\, T_0 + W]} \text{gap}_t, \tag{2}$$

the mean gap over the post-window W. The ATT is preferred over the peak or terminal gap because it integrates over the whole path including reversion, the object that reserve sizing, pass-through, and fiscal buffering act on, and it is less sensitive to single-day jumps from liquidity or futures-roll mechanics.

3.3 Identifying assumptions

The factor model makes the assumptions transparent.

1. **Pre-period fit / common-factor spanning.** The donor basket must reproduce Brent's factor loadings, so that pre-event tracking is good and is driven by shared F_t, not coincidence. This is checked by pre-period RMSPE, walk-forward cross-validation, and moment matching.

2. **Donor cleanliness (SUTVA).** No donor carries the treatment: γ_j must not load on o_t, and donor j must not be hit through a channel specific to the focal event. This is the no-interference condition; it is defended by an a-priori audit and confirmed by model-agnostic break tests at T_0.

3. **Regime stability.** The factor loadings λ, γ_j must be stable between the pre-window and the projection window, so the basket that tracked Brent before T_0 remains its valid counterfactual after. This is checked by distance-correlation and structural-break diagnostics, and, across events, by the cross-event weight transfer.

3.4 The sign of contamination bias

A virtue of the factor representation is that it signs the bias from imperfect donor cleanliness exactly. With $\delta_{jt} = X_{jt} - X_{jt}^N$ the event-induced component of donor j,

$$\hat{\tau}_t - \tau_t \;=\; Y_t^N - \hat{Y}_t^N \;=\; -\sum_j w_j\, \delta_{jt}. \tag{3}$$

Table 1: The five-estimator ensemble. Each contributes a distinct inductive bias; the headline is the cross-model median gap.

Estimator	Inductive bias	Reference
Convex SCM	Non-negative donor weights summing to one; synthetic inside the donor convex hull	Abadie et al. (2010)
Augmented SCM	Convex SCM plus ridge bias-correction; limited extrapolation outside the hull	Ben-Michael et al. (2021)
Elastic net	Signed, sparse+ridge regression; lasso term drops donors entirely	Zou and Hastie (2005); Doudchenko and Imbens (2016)
Gradient boosting (XGBoost)	Nonlinear, tree-based; captures interactions; cannot extrapolate beyond training range	Friedman (2001); Chen and Guestrin (2016)
Bayesian ridge	Bayesian linear regression on donor levels (regression component of BSTS); conjugate Gaussian–Gamma prior	Brodersen et al. (2015)

The bias is *minus* the weight-averaged donor contamination, an empirical quantity, not an assumed sign. If contaminated donors are pushed *up* (oil-linked terms-of-trade, inflation→rates), the synthetic is pulled up and the ATT is *under*-estimated (conservative); if pushed *down* (demand crowd-out, risk-off), the ATT is over-estimated. Equation (3) turns "the donors might be affected by the war" from an informal worry into a quantity we can measure and sign (Section 6). It also separates *contemporaneous* contamination (at T_0, removed by the audit) from *delayed* contamination (accruing over the post-window, controlled by the post-window length): delayed contamination enters through the projection, biases the gap toward zero, and grows with horizon, so a gap that *declines* with horizon is its signature and the short-horizon gap is the least-contaminated estimate.

3.5 Why an ensemble

The counterfactual's functional form is unknown: equation (1) need not be linear in the donors, and the convex-hull restriction of classical SCM may bind. Rather than commit to one specification, we proxy the counterfactual with five estimators whose inductive biases bracket the plausible forms (Table 1). If the estimated premium survives across models with different biases (a simplex-constrained average, a ridge-extrapolating correction, a sparse signed regression, a nonlinear tree ensemble, and a Bayesian linear model), it is far more likely to reflect a real effect than a single-model artefact. The headline is the cross-model *median* (robust to one model misbehaving) and the inter-quartile range is the model-uncertainty band, complementary to each model's own within-model interval.

A note on labelling: the fifth model is a Bayesian *ridge* regression, not full Bayesian structural time series. It implements only the regression-on-donors component $\beta' x_t$ of Brodersen et al. (2015) and omits the trend, seasonal, and spike-and-slab selection components; at our sample size, with no strong weekly seasonality, the regression term carries most of the signal, but the model is best read as a Bayesian-flavoured diversifier rather than a state-space causal-impact estimator (Appendix B).

4 Data

4.1 Outcome and sources

The outcome is the daily *Europe Brent Spot Price FOB* (EIA series RBRTEd), in USD per barrel, downloaded directly from the U.S. Energy Information Administration (U.S. Energy Information Administration, 2026). The donor candidates are 32 macro-financial series from Yahoo Finance (futures, ETFs, FX, rates, volatility), updated incrementally so the panel keeps pace with new data. The Caldara and Iacoviello (2022) daily Geopolitical Risk index enters only as a descriptive overlay, never as a donor, because it spikes endogenously to the events we study. Table 2 summarises the sources; all are public and redistributable, and the panel is rebuilt from source on each run so the analysis is reproducible against the latest available data (here, through 15 June 2026 for Brent, the binding constraint given the EIA's roughly one-week publication lag, and 23 June 2026 for the donors).

Table 2: Data sources. All series are daily and publicly redistributable.

Object	Source	Contents
Brent spot (outcome)	EIA, series RBRTEd	Europe Brent Spot FOB, USD/bbl, 1987–present
Donor candidates (32)	Yahoo Finance	Metals, agriculturals, equities, FX, rates, credit, volatility, crypto
GPR index (overlay)	Caldara–Iacoviello	Daily Geopolitical Risk index, 1985–present (not a donor)

4.2 The donor pool and the SUTVA audit

Donor selection is central to identification (Section 3): a series qualifies only if it (i) co-moves with Brent through common global factors, (ii) is *not* physically routed through the disruption being attributed, (iii) is liquid and daily-traded, and (iv) has sufficient pre-history. Criterion (ii) is the SUTVA (no-interference) condition and is enforced by a per-event audit that classifies each donor as Clean, Mild, or Heavy. For **Russia 2022**, 13 of 32 donors are flagged through identifiable channels: Russia/Ukraine grain exports (Wheat, Corn), palladium supply concentration, the EU energy-crisis pass-through to EUR and the dollar basket, and a sanctions-evasion narrative for Bitcoin. For **Hormuz 2026**, only Cotton is flagged (the indirect oil→polyester substitution channel); no donor is physically routed through the Strait.

To make the two events directly comparable, and to enable the cross-event weight transfer, we use a single shared 19-donor pool: the Russia strict-clean set, which is a subset of the Hormuz clean set. One donor, VIX, is borderline: a full-history Bai–Perron test (Bai and Perron, 1998) flags a single break in its relationship with Brent on 2022-01-07 ($\sim$7 weeks before the invasion). We treat that as a loading-stability signal rather than a SUTVA violation, because it does not reproduce when the test is run on the analysis window alone, and a sensitivity check shows that keeping or dropping VIX moves the ensemble premium by less than 1 percentage point. We therefore retain VIX, which keeps the volatility factor represented and matches the specification used in our milestone presentation. The energy complex (WTI, refined products, petrocurrencies) is excluded by construction: it has the highest raw correlation with Brent but loads on the very oil-specific shock we measure. Table 3 lists the 19-donor pool by factor category.

Table 3: The shared 19-donor pool, by factor category. The full 32-donor pool adds Russia-contaminated series (e.g. Wheat, Corn, Palladium, EUR, DXY, BTC, Copper, Iron Ore, US10Y) used only in the appendix robustness pool and as a contamination diagnostic.

Category	Donors (shared pool)
Metals (3)	Silver, Platinum, Gold
Agriculturals (3)	Coffee, Sugar, Live Cattle
Equities (2)	S&P 500, Nikkei 225
Foreign exchange (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN
Rates & credit (2)	TLT (long Treasuries), HYG (high-yield credit)
Volatility (1)	VIX (implied S&P 500 volatility)

4.3 Sample windows and summary statistics

Each event has a pre-window (for fitting) and a post-window (for projecting the counterfactual). Pre-windows are ∼20 months, long enough for factor identification yet short enough to exclude acute Brent-specific contamination. Following Abadie (2021), the Russia pre-window starts 2020-07-01, deliberately *after* the March–May 2020 cluster (OPEC+ collapse, WTI-negative, acute COVID) that would otherwise teach the synthetic an abnormal factor loading. The Russia post-window is truncated at 2022-09-30, the last business day before OPEC+'s October 2022 supply cut (a Brent-specific policy shock that would confound the invasion premium). The Hormuz post-window extends to the latest available data (2026-06-15). Table 4 reports summary statistics; Figure 1 plots the full Brent series with both events.

Table 4: Summary statistics of Brent spot (USD/barrel) over each event sample. Pre/post split at T_0. Russia: T_0=2022-02-24; Hormuz: T_0=2026-02-28.

Event	Segment	n (days)	Mean	SD	Min	Max
Russia 2022	Full sample	572	75.95	24.76	36.33	133.18
	Pre-window	421	64.26	16.39	36.33	101.66
	Post-window	151	108.54	11.37	82.55	133.18
Hormuz 2026	Full sample	515	76.95	14.30	59.93	138.21
	Pre-window	443	72.08	6.58	59.93	88.66
	Post-window	72	106.92	12.31	77.24	138.21

The two pre-windows differ in a way that matters later. The Russia pre-period is volatile (SD \$16.4, spanning the COVID recovery and the 2021 reflation rally); the Hormuz pre-period is much calmer (SD \$6.6). This difference drives much of the contrast in the two events' inference: a volatile pre-period widens the placebo distribution and weakens Russia's permutation test, even though its magnitude is well-documented.

5 Empirical Strategy and Validation

5.1 Estimators

All five estimators (Table 1) take the same input (log donor levels) and target log Brent over the pre-window, then project the post-window counterfactual. *Convex SCM* solves the classic nested optimisation: non-negative weights summing to one, chosen to minimise pre-period RMSPE (Abadie et al., 2010). *Augmented SCM* adds a ridge penalty that corrects for imperfect pre-fit and allows limited extrapolation (Ben-Michael et al., 2021). The *elastic net* relaxes the simplex to signed, sparse coefficients (Zou and Hastie, 2005; Doudchenko and Imbens,

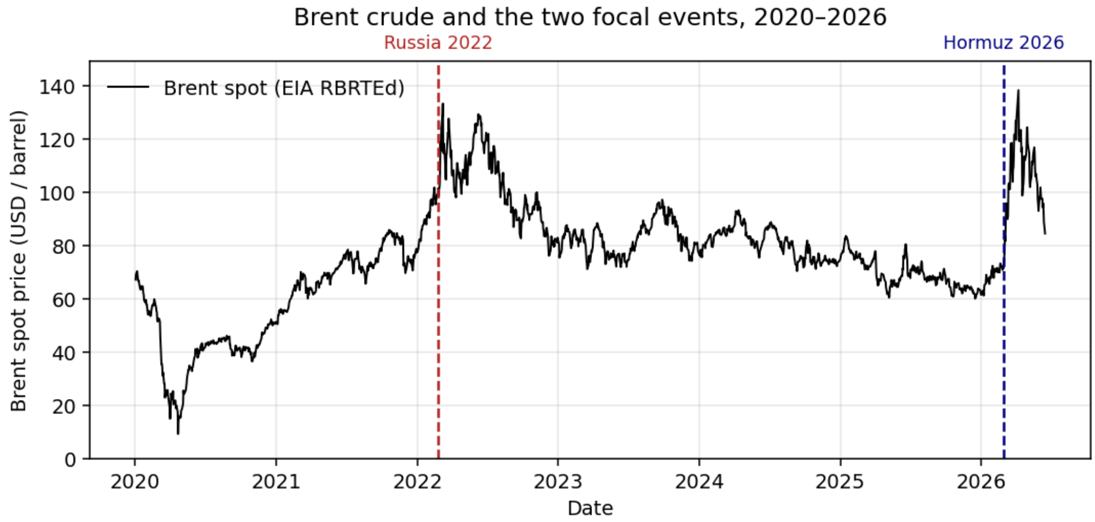

Figure 1: Brent crude spot price, 2020–2026, with the two focal event dates. Brent roughly doubled after both shocks from a similar ∼$75 base, but from very different pre-event trajectories: a steep 2020–22 recovery rally before Russia, and a mild net decline through 2024–25 before Hormuz. This trajectory asymmetry is what makes a naïve trend-extrapolation counterfactual unreliable (Section 6).

2016). *XGBoost* fits shallow gradient-boosted trees (Chen and Guestrin, 2016), deliberately constrained (depth-2 stumps, strong regularisation, row/column subsampling) for the small-sample regime. *Bayesian ridge* is a conjugate Bayesian linear regression on donor levels, the regression component of BSTS (Brodersen et al., 2015). Hyperparameters are tuned over small grids (3–9 configurations) to bound the selection bias of Cawley and Talbot (2010); the convex SCM has no tuned hyperparameter. We implement the pipeline in Python: the convex and augmented SCM solvers are written directly on top of `numpy`/`scipy`, the elastic net and Bayesian ridge come from `scikit-learn`, gradient boosting from `xgboost`, and the ARIMA baselines and structural-break statistics from `statsmodels` plus our own implementations of the Bai–Perron and Harvey–Leybourne–Taylor tests.

5.2 Donor cleanliness battery

Before any model is fit, each donor's own return series is tested for a mean shift at T_0 using a permutation form of the known-break test of Chow (1960) (rather than the unknown-break test of Andrews (1993), since the date is known), together with a Wilcoxon event-window test (Brown and Warner, 1985); a donor is flagged if either rejects after Benjamini–Hochberg FDR control (Benjamini and Hochberg, 1995). Regime stability is checked with distance correlation across pre-period thirds (Székely et al., 2007) and with Bai–Perron break dating (Bai and Perron, 1998, 2003); an I(0)/I(1)-robust trend-break test (Harvey et al., 2009; Kwiatkowski et al., 1992) confirms no donor's own trend breaks at either T_0. Empirically, for Hormuz 31 of 32 donors agree between audit and tests (only Cotton differs); for Russia the tests confirm the broad picture and add one flag (CNY) attributable to concurrent China dynamics, not the war.

5.3 Validation battery and its logic

A credible SCM estimate must clear three levels: pre-event fit, donor identification, and inference (Abadie, 2021). We run, per model and event: (a) expanding-window walk-forward cross-validation for out-of-sample fit (Hyndman and Athanasopoulos, 2018; Bergmeir and Benítez, 2012); (b) the in-space placebo, refitting each donor as a placebo treated unit and ranking Brent's post/pre RMSPE ratio against the resulting distribution (Abadie et al., 2010), reading the permutation p-value with the symmetry caveat of Hahn and Shi (2017); (c) the in-time placebo at a fake T_0 six months early, formalised as the mixed placebo test of Chen and Yan (2023); (d) leave-one-donor-out; (e) the cross-event weight transfer; (f) a deliberately contaminated set of naïve univariate counterfactuals (random walk, drift, linear trend, frozen ARIMA) as a floor; and (g) the signed contamination-bias diagnostic of equation (3). Following Roth (2022); Rambachan and Roth (2023), pre-period diagnostics are reported and interpreted rather than turned into pass/fail thresholds; flagged models are *not* auto-excluded, because the cross-model IQR is itself the model-uncertainty band.

The headline per event is the cross-model *median* gap, with the IQR as the band. Where a model's pre-period gap series trends, we report a drift adjustment: a slope of $\hat{\beta}$ %/yr over a post-window of L years contributes $\hat{\beta}L$ percentage points independent of treatment, so the drift-adjusted gap subtracts it.

6 Results

6.1 Pre-event fit

Table 5 reports walk-forward CV. For Hormuz, four of five models clear the heuristic flag (validation/training RMSE ratio below 2); only Bayesian ridge, at 2.28, is flagged. For Russia, two of five clear it (convex SCM and elastic net); with VIX kept in the pool the augmented SCM rises just past the threshold (2.06), and XGBoost (3.24) and Bayesian ridge (3.09) fail more clearly. The flags are not exclusions: they identify which models to read with caution, and the most flagged model (Bayesian ridge) is the one whose headline gap is the ensemble outlier.

Table 5: Walk-forward cross-validation (5 expanding folds, 20-day horizons). A validation/training RMSE ratio above 2 is a heuristic overfit flag, not an exclusion rule.

Event	Model	train RMSE	val RMSE	val/train	Flag
Russia	Convex SCM	0.113	0.141	1.25	OK
Russia	Augmented SCM	0.046	0.095	2.06	overfit
Russia	Elastic net	0.051	0.102	1.99	OK
Russia	XGBoost	0.039	0.127	3.24	overfit
Russia	Bayesian ridge	0.034	0.105	3.09	overfit
Hormuz	Convex SCM	0.060	0.083	1.40	OK
Hormuz	Augmented SCM	0.049	0.075	1.54	OK
Hormuz	Elastic net	0.052	0.079	1.51	OK
Hormuz	XGBoost	0.044	0.082	1.84	OK
Hormuz	Bayesian ridge	0.034	0.078	2.28	overfit

6.2 Headline estimates

Table 6 reports the per-model post-window gap, the drift contribution, and the drift-adjusted gap, for both events; Figure 2 plots the observed and synthetic paths. **Russia 2022** yields an ensemble-median premium of 22.0 % over the seven-month window (IQR 10.9 % to 30.8 %), with an implied counterfactual mean of \$88.6/bbl against an actual \$108.5. This is consistent with the documented \$95→\$130 move and its partial reversion by September. **Hormuz 2026** yields an ensemble-median premium of 55.9 % over the window to mid-June (IQR 51.2 % to 56.6 %), with an actual post-window mean of \$106.9 against an implied counterfactual of ∼\$68.6.

The Hormuz IQR is far tighter than Russia's (~5 vs ~20 pp), reflecting the calmer Hormuz pre-period.

Table 6: Headline post-event gaps (%). Drift-adjusted = raw − drift contribution. Ensemble = cross-model median; band = IQR (Q25–Q75).

	Russia 2022 (7 mo)			Hormuz 2026 (~3.5 mo)		
Model	Raw	Drift	Adj.	Raw	Drift	Adj.
Convex SCM	33.6	+5.2	28.4	55.9	−1.4	57.3
Augmented SCM	10.9	+0.3	10.6	56.8	−0.4	57.1
Elastic net	22.0	+0.7	21.2	56.6	−0.8	57.4
XGBoost	30.8	+1.8	28.9	51.2	−1.5	52.7
Bayesian ridge	−9.0	+0.1	−9.1	43.0	−0.0	43.0
Ensemble median	**22.0**	n/a	**21.2**	**55.9**	n/a	**57.1**
IQR	**[10.9, 30.8]**	n/a	n/a	**[51.2, 56.6]**	n/a	n/a

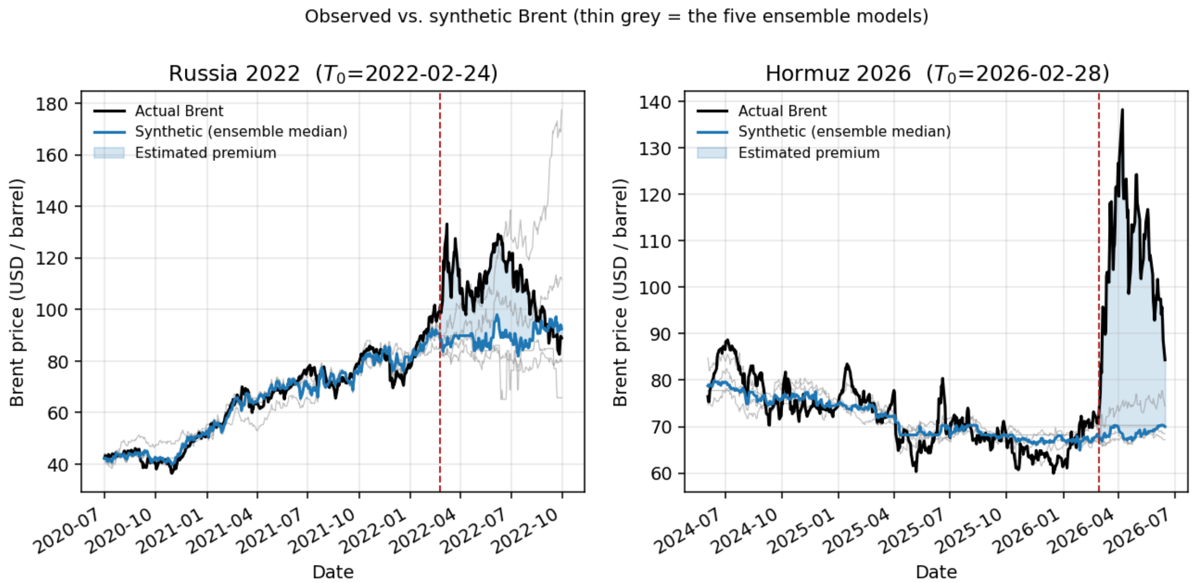

Figure 2: Observed Brent (black) vs. the synthetic counterfactual. The thick blue line is the ensemble-median synthetic; thin grey lines are the five individual models; the shaded region after T_0 (dashed red) is the estimated premium. The synthetics track Brent closely pre-event and decouple sharply afterward.

Economic reading of the model spread. The cross-model dispersion has an economic explanation rather than being noise. Convex SCM, confined to the simplex, leans on the most stable donors (the FX block) as a "trend baseline" and adds soft-commodity deltas (Coffee, Sugar for Russia). The Bayesian ridge, with unconstrained Gaussian coefficients, loads heavily and with large opposing signs on currencies; for Russia those currencies were pushed around by the simultaneous, historically large Fed tightening cycle, a macro factor orthogonal to the invasion. That is why Bayesian ridge is the Russia outlier (a negative full-window gap) and why it earns the highest walk-forward overfit flag: it is the expected behaviour of an unregularised linear model in a pre-period with a strong non-event macro driver, and it is the reason the ensemble uses the *median* rather than the mean. For Hormuz, whose calmer pre-period lacks an analogous confound, the five models agree to within ~5 pp.

6.3 Inference: the in-space placebo

Table 7 and Figure 3 report the in-space placebo. For **Hormuz**, Brent ranks *first* among all placebo units under every model and both pools, with a post/pre RMSPE ratio (6.9–10.2) above every donor's, so the permutation p-value sits at the attainable floor (0.050 in the 19-donor pool, 0.030 in the 32-donor pool). This is the strongest signal the test can give at this sample size. For **Russia**, no model reaches the floor (XGBoost comes closest at $p = 0.20$ in the 19-donor pool, 0.15 in the 32-donor pool); the volatile 2020–22 pre-period gives donors large pre-period RMSPE too, widening the placebo distribution. This is a known low-power property of the in-space placebo at volatile pre-periods (Abadie et al., 2010), not evidence of a small effect. Russia's magnitude is independently documented and externally validated below.

Table 7: In-space placebo permutation p-values and Brent's post/pre RMSPE ratio (19-donor pool). Floor $= 1/(N{+}1)$: 0.050 (19-donor), 0.030 (32-donor).

Event	Model	p (19-donor)	p (32-donor)	Brent ratio
Russia	Convex SCM	0.50	0.50	2.75
Russia	Augmented SCM	0.60	0.47	3.51
Russia	Elastic net	0.40	0.41	4.08
Russia	XGBoost	0.20	0.15	7.75
Russia	Bayesian ridge	0.60	0.53	5.88
Hormuz	Convex SCM	**0.050**	**0.029**	6.93
Hormuz	Augmented SCM	**0.050**	**0.029**	8.45
Hormuz	Elastic net	**0.050**	**0.029**	8.53
Hormuz	XGBoost	**0.050**	**0.029**	10.19
Hormuz	Bayesian ridge	**0.050**	**0.029**	10.24

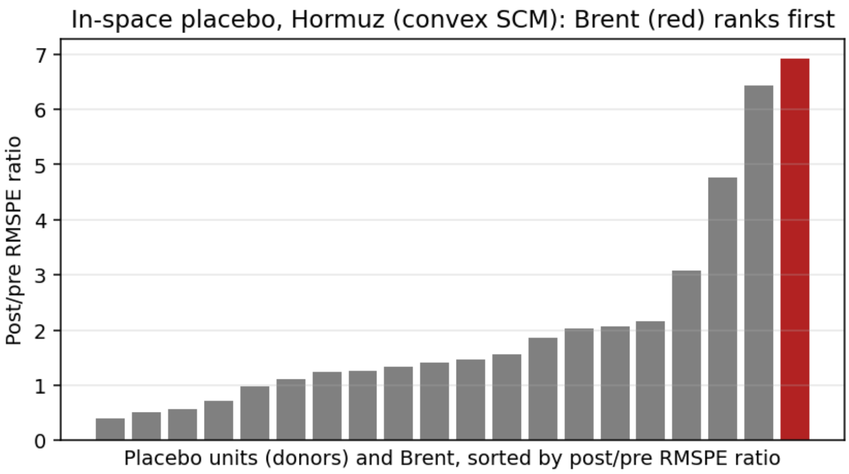

Figure 3: In-space placebo for Hormuz (convex SCM). Each bar is one unit's post/pre RMSPE ratio; Brent (red) is the largest, ranking first among all 19 placebo units, so the permutation test rejects at the attainable floor.

6.4 Robustness

Russia magnitude / external validation. Because Russia is a known event, we validate its magnitude against the EIA's last two *pre-invasion* Short-Term Energy Outlook forecasts (issued before 24 February 2022), which are an independent structural counterfactual sharing no information path with the SCM. Over the identical 151-day window, the ensemble-median synthetic mean ($88.6) sits within $3.6 of the STEO Feb-2022 forecast ($84.9); three of five models bracket the STEO anchor to within ±$4. Two independent methods agreeing on the counterfactual is strong evidence the SCM is not producing an implausible path.

Post-window horizon sensitivity. Figure 4 reports the gap at 1/2/3/6-month and full horizons. The two events show opposite, diagnostic profiles. Russia *declines* with horizon (31% → 22%): the signature of delayed contamination, as second-round war channels switch on through 2022 and pull the synthetic up, so the full-window figure is conservative and the least-contaminated one-month estimate is ∼31%. Hormuz *rises then eases* (46% → 59% → 56%): the chokepoint premium builds in and holds, the opposite of attenuation; the slight easing to 56% at the full horizon reflects Brent's partial mid-June pullback, captured by extending the window to the latest data. That the same diagnostic declines for Russia and rises for Hormuz shows it is not a mechanical artefact.

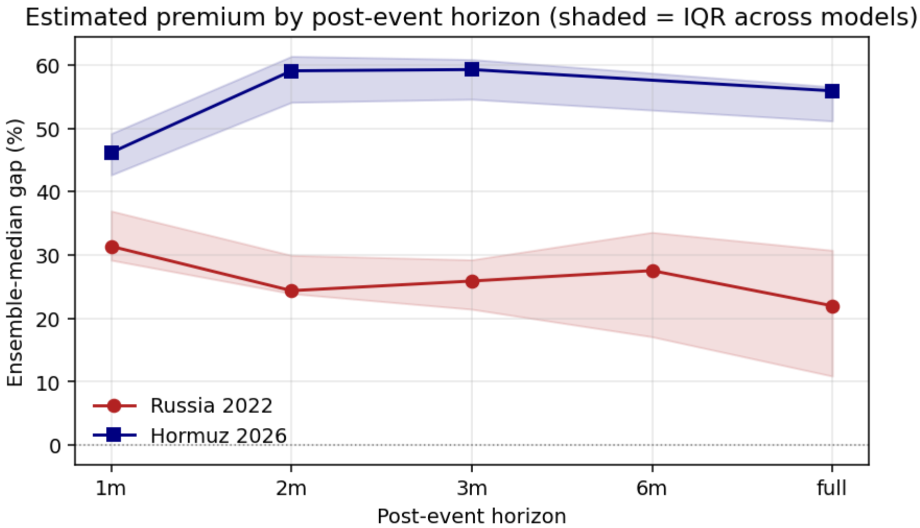

Figure 4: Ensemble-median gap by post-event horizon (shaded = IQR across models). Russia attenuates with horizon (delayed-contamination signature); Hormuz rises then plateaus (the premium builds in and holds).

Leave-one-donor-out. Dropping each weighted donor in turn leaves the Hormuz estimate tight around its baseline (convex 51.2–59.3, ASCM 54.6–60.5, elastic 55.2–62.2, XGBoost 51.6–57.7%); no single donor drives the result. Russia is wider (e.g. convex 32.8–59.6%), consistent with its volatile pre-period and the simplex's reliance on a few soft-commodity donors.

Cross-event weight transfer. Applying Russia-fitted weights to the Hormuz window (Table 8), convex SCM is the only model that transfers to a clearly *positive* Hormuz premium (25.3%) with a degraded but non-pathological pre-fit (pre-RMSPE 5× independent); the augmented SCM transfers to roughly zero (−0.8%, 7×), and the signed-regression, tree, and prior-dominated models go strongly negative with pre-RMSPE 14–280× independent. We therefore read the transfer as weak, convex-only corroboration that *some* positive premium survives a change of regime, not as confirmation of its level: the headline magnitude rests on the within-event ensemble. Once the window is extended through Brent's June pullback, the regime gap between the 2020–22 and 2024–26 factor structures is evidently wider than over the shorter window.

Table 8: Cross-event weight transfer: Russia-fitted weights applied to Hormuz. Independent and transferred pre-period RMSPE in log units.

Model	Indep. gap (%)	Transf. gap (%)	Indep. RMSPE	Transf. RMSPE
Convex SCM	55.9	25.3	0.066	0.336
Augmented SCM	56.8	−0.8	0.066	0.467
Elastic net	56.6	−68.6	0.054	1.513
XGBoost	51.2	−44.1	0.053	0.928
Bayesian ridge	43.0	−100.0	0.037	18.567

Naïve counterfactual floor. Against donor-free univariate counterfactuals (Table 9), the SCM adds value, and the naïve methods' bias *flips sign across events* purely because the pre-window trend sign differs. A linear trend extrapolates Russia's recovery rally upward and returns an implausible *negative* full-window gap (−4%), but extrapolates Hormuz's mild downtrend and *inflates* the gap to 71%. The trend-free random-walk null gives Russia +8.7% and Hormuz +48.9%, with the SCM ∼13 and ∼7 pp above respectively. The SCM, anchored to donor co-movement rather than Brent's own momentum, is far more stable across the two regimes, which is the concrete demonstration of why a donor-based counterfactual is necessary.

Table 9: SCM vs. naïve univariate counterfactuals, full-window gap (%). RW flat/drift: random-walk carry / with drift; Lin. trend: linear pre-trend extrapolation; ARIMA rw/mr: frozen AR(1) forecasts (random-walk / mean-reverting).

Event	SCM median	RW flat	RW drift	Lin. trend	ARIMA rw	ARIMA mr
Russia	22.0	8.7	−6.9	−4.2	9.2	26.5
Hormuz	55.9	48.9	49.8	70.9	48.1	48.7

Sign of the contamination bias. Using equation (3) with a Brent-free common factor, we sign the bias per (linear) model. For **Hormuz** the full-window bias is small and *negative* (conservative) for all four linear models (−1.3 to −7.6%), so the large premium is not a contamination artefact. For **Russia** the full-window bias is large and *positive* (overestimate, +13 to +21% for three of four models), concentrated in low-reliability soft commodities whose 2022 idiosyncratic drift the convex/elastic fits leaned on; following the reliability logic of Ham and Miratrix (2024), splitting donors by pre-window factor R^2 shows the bias sits mostly in the low-R^2 (unreliable) donors. The asymmetric conclusion is that contamination is conservative for the focal Hormuz estimate but inflates the Russia *full-window* number; Russia's magnitude validation therefore rests on the early/peak window (where the \$95→\$130 move is recovered), not on the attenuated full-window average.

6.5 Interpretation

Our estimate places the 2026 Hormuz chokepoint premium at roughly 56 % of Brent's no-event level over the first ∼3.5 months. The number holds up across all five estimators and across the checks we ran: the in-space placebo reaches its floor, leave-one-out stays tight, the contamination bias is conservative, and the SCM clears the naïve baselines. The cross-event transfer is the weakest of the checks: only convex SCM transfers to a clearly positive premium, so we lean on it for the sign rather than the level. We read the estimate as a complement to, not a competitor of, the structural scenario estimates from energy institutions: those answer "what price does an assumed barrel shortfall imply," while the SCM answers "what price gap does the observed cross-asset co-movement imply." For the policy levers of Section 2 (reserve releases, monetary pass-through, fiscal buffers), the integrated, path-level premium is the relevant input, and the estimate here provides it with an auditable identification argument.

7 Conclusion

We estimated the Brent-price effect of the 2026 Strait of Hormuz disruption by building a counterfactual with an ensemble synthetic control, validating the design first on the known Russia 2022 event. Over the window to mid-June 2026 the ensemble-median premium is ∼56% (IQR 51–57%), large and positive under every estimator, and it survives the in-space placebo (Brent first among all units, p at the floor), leave-one-out, the contamination-bias sign check, and the naïve-baseline comparison; the cross-event transfer gives weaker, convex-only support for a positive premium. On the calibration event the same machinery recovers a magnitude consistent with the documented \$95→\$130 move and within \$3.6/bbl of the EIA's pre-invasion structural forecast.

The estimate's value is as a reduced-form, data-driven complement to structural scenario analysis, with an identification argument a reviewer can audit donor by donor. Several limitations qualify it. The Hormuz post-window is short (∼72 observations), so every inference test is under-powered and the placebo can only reach its floor; the convex synthetic under-represents Brent's volatility, which the augmented SCM corrects; the cross-event transfer degrades once the window is extended through Brent's June pullback; the Bayesian-ridge model is a regression proxy for BSTS, not the full state-space model; and the estimand is a reduced-form total effect, not a supply/demand/risk decomposition. Natural extensions are a longer post-window as data accrue, per-donor delayed-contamination onset dating to sharpen the post-window choice, and a full BSTS implementation for autocorrelation-aware credible intervals. None of these changes the central finding: the observed co-movement of global markets with Brent implies that the 2026 Hormuz disruption added a large, statistically distinguishable premium to crude, on the order of half its no-event price.

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A Robustness grid

The headline uses the preferred specification (shared 19-donor pool, preferred pre-window, ensemble median). The full grid varies (i) the pre-window (preferred, extended, and narrow specifications); (ii) the donor pool (shared 19, permissive 26 for Russia, full 32); and (iii) the model (each of the five plus the median). For Russia the donor-pool sweep is itself a diagnostic: the gap is expected to *shrink* monotonically from the clean 19-donor pool to the contaminated 32-donor pool, because Russia-treated donors get inflated by the same shock and pull the synthetic up. If this monotonic pattern holds, the SUTVA-driven exclusion logic is empirically supported.

B Bayesian ridge and full BSTS

The fifth ensemble member implements only the regression-on-donors term $\beta' x_t$ of the Bayesian structural time-series model of Brodersen et al. (2015), with a conjugate Gaussian–Gamma prior. It omits the local-linear *trend*, the *seasonal* component, the *spike-and-slab* donor selection, and the autocorrelation-aware posterior over the full state trajectory. At our sample size, with no strong weekly seasonality in daily oil prices, the regression term carries most of the signal, so the proxy is an adequate Bayesian-flavoured diversifier; but its credible intervals are Gaussian and ignore autocorrelation, and its donor "importances" are standardised-coefficient magnitudes, not true posterior inclusion probabilities. We therefore call it a Bayesian ridge regression rather than a BSTS, and read its placebo p-values as rank statistics rather than formal probabilities (Table 7). A full BSTS implementation is the natural route to autocorrelation-aware credible intervals in a future iteration.

C Structural-break tests

Two break tests support donor screening. *Bai–Perron* (Bai and Perron, 1998, 2003) dates breaks in the donor–Brent weekly-return relationship $r_t^{\text{donor}} = \alpha_s + \beta_s r_t^{\text{Brent}} + \varepsilon_t$ by globally minimising segment SSR via dynamic programming, with the break count chosen by BIC. Only VIX shows a break strictly inside a pre-window (2022-01-07). We treat the break tests as informative rather than automatically disqualifying: the VIX break is a loading-stability signal, it does not reproduce on the analysis window alone, and the in/out sensitivity is below 1 pp, so VIX is retained (Section 4). The relationship specification is Brent-based, so a break *at* T_0 is ambiguous; the complementary Harvey et al. (2009) trend-break test resolves this by testing the donor's *own* log-level trend with a statistic whose asymptotic size is invariant to whether the series is I(0) or I(1): it weights an I(0)-optimal and an I(1)-optimal sup-t statistic by KPSS stationarity weights (Kwiatkowski et al., 1992). Empirically all 32 donors are classified I(1)-like and none rejects a trend break within ± 8 weeks of either T_0, so no retained donor was itself trend-treated by either event.